

5.2 Options Contracts



Guiding Questions

1. How does an option differ from a futures contract in terms of obligation and payoff?
2. What creates intrinsic value vs time value in option pricing?
3. How can a producer use put options to protect revenue while keeping upside?
4. What are the main risks of selling options versus buying them?

What will you learn?

- The definition of call and put options and their payoff profiles.
- The four key drivers of option prices (spot price, strike, time, volatility).
- How protective puts create downside insurance for commodity producers.
- Covered calls as income generation for hedgers with inventory.

5.2.1 The Anatomy of an Options Contract

An options contract gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a fixed strike price (K) by or on a specific expiration date.

The option seller (writer) receives a premium upfront but has the obligation if the buyer exercises.

Two Fundamental Types

Type	Right Granted	Used When	Payoff Profile
Call Option	Right to BUY at strike price K	Expect prices to RISE	Unlimited gain, max loss = premium
Put Option	Right to SELL at strike price K	Expect prices to FALL	Limited gain (to zero), max loss = premium

Key Difference from Futures: Options have no margin calls for buyers—maximum loss is the premium paid upfront. Sellers face potentially unlimited risk (calls) or substantial risk (puts).

☐ Watch

Options



You now understand what functions are and why analysts rely on them.

Next, you'll **practice using and modifying functions and** take your first step towards writing your own (a key milestone in developing analytical confidence).

5.2.2 Key Components and Terminology

5.2.2.1. Strike Price (K)

The fixed price at which the option can be exercised. S is the actual market price (spot price) when the option can be exercised.

5.2.2.2. Premium

The market price of the option, determined by supply/demand. Composed of:

$$\text{Premium} = \text{Intrinsic Value} + \text{Time Value}$$

Intrinsic Value: Immediate exercise value (max of zero)

Call: $\text{Max}(S - K, 0)$; if $S > K$ then profit can be made, otherwise walk away and payoff will be 0 but the premium paid in advance is also lost (nonrefundable).

Put: $\text{Max}(K - S, 0)$

Time Value: Extra value due to time remaining and volatility

5.2.2.3. Moneyness

In-the-Money (ITM): Has intrinsic value

At-the-Money (ATM): Strike $\approx$ current spot price

Out-of-the-Money (OTM): No intrinsic value

5.2.4. Expiration

Options expire worthless if not exercised. American options can be exercised anytime; European only at expiry.

5.2.3 Option Payoff Profiles

Call option:

The graph below involves an option consisting of 100 shares, strike price (aka exercise price) $K = \$50$ and per share when the premium (aka price of the option) is \$5 per share.

In the left panel of the graph below, when the actual price of the stock (aka the spot price) $S = 50$ there is a loss of \$ 500 ($\$5 \times 100 = 500$), because the premium paid in advance is nonrefundable. But when $S > K$, there is a profit which will be higher for higher values of S . The maximum loss in

the case of call option is the premium paid but the profit can be unlimited as you can see on the right hand side of the left panel.

Put Option:

In the of the put option, shown in the right panel of the above figure, when $S = 50$ there is a loss of \$ 500 ($\$5 \times 100 = 500$). When $S < K$, there is a profit which will be higher for lower values of S . When $S < K$, the maximum loss is still \$500 as the buyer of this put option has the right to walk away and not exercise this option. The above graph highlights the fact that maximum loss on an option contract is limited to the premium paid. Of course, the option writer (i.e., the broker who sold these options can have huge losses and the maximum gain is the premium received. In real life, option seller/writer create and sell many options – they lose money on a few and may money on several (just like the insurance market where normally only a few people file large insurance claims and most people simply pay the premium).

5.2.4 The Four Drivers of Option Prices (The Greeks)

1. Spot Price (S) → Delta (Δ)

Higher spot price increases call values, decreases put values. Delta measures sensitivity (0 to 1 for calls, -1 to 0 for puts).

2. Strike Price (K) → Reverse Delta

Higher strike favours puts, hurts calls.

3. Time to Expiry → Theta (Θ)

Longer time increases both call and put values (more chance to go ITM). Time decay accelerates as expiry approaches.

4. Volatility (σ) → Vega (v)

Higher volatility massively increases option premiums (more chance of extreme moves).

Example: A wheat call option's premium might jump 50% if weather forecasts predict drought.

5.2.5 Clearinghouse and Standardization (Same as Futures)

Options trade on exchanges like ASX 24 Options or CBOE, cleared by:

ASX Clear (Australia)

OCC (Options Clearing Corporation) (USA) - <https://www.theocc.com/> 
[\(https://www.theocc.com/\)](https://www.theocc.com/)

Standard contract sizes ensure liquidity:

Equity options: 100 shares

Commodity options: Match underlying futures size

The clearinghouse guarantees performance exactly as in futures markets.

5.2.6 Worked Example: Wheat Producer Using Put Options

Situation: A local wheat farmer expects 10,000 tonnes harvest in 3 months.

Current spot: \$600/tonne. Fears price drop but wants upside if prices rise.

Strategy: Buy put options with strike $K = \$600$, premium = \$25/tonne.

Case A – Price Falls to \$500

Physical sale: $\$500 \times 10,000 = \$5,000,000$

Put payoff: Exercise at \$600 $\rightarrow (\$600 - \$500) \times 10,000 = +\$1,000,000$

Premium cost: $\$25 \times 10,000 = -\$250,000$

Net revenue: \$5,750,000 (effective price \$575/tonne)

Case B – Price Rises to \$650

Physical sale: $\$650 \times 10,000 = \$6,500,000$

Put expires worthless (do not exercise)

Premium cost: $-\$250,000$

Net revenue: \$6,250,000 (effective price \$625/tonne)

Case C – Price Stays at \$600

Physical sale: \$6,000,000

Put expires worthless

Net: \$5,750,000 (cost of insurance)

Interpretation: The put provided downside protection at \$575 while allowing unlimited upside above \$625. Compare to futures (\$600 fixed).

5.2.7 Risk Comparison: Futures vs Options

Aspect	Futures (Short Hedge)	Put Options (Protective Put)
Downside Protection	Perfect (\$600)	Good (\$575 after premium)

Upside Potential	None (capped \$600)	Unlimited above \$625
Max Loss	Unlimited margin calls	Premium only (\$25/tonne)
Cash Flow	Daily MTM volatility	One-time premium
Flexibility	Binding obligation	Right, not obligation

Daily Mark-to-Market (MTM) volatility refers to the daily fluctuation in a portfolio's value due to price changes, resulting in realized cash settlements or margin adjustments. It is measured by the standard deviation of daily price changes, directly affecting liquidity and risk management.

Conclusion: Options cost money upfront but preserve upside and eliminate margin risk.

5.2.8 Hedging Equities with Options (Buying Insurance)

An investor who wants protection against a market decline but still wishes to participate in a market rally can use options instead of futures.

- **Single Stock:** To protect an individual stock, the investor buys an **equity put option** on that specific company.
- **S&P 500:** To protect a broad portfolio, the investor buys an **S&P 500 index put option**.

The Mechanics & Trade-off:

This strategy creates an asymmetric hedge that functions exactly like an insurance policy. The investor must pay an upfront cash fee (the option premium). If the stock or the S&P 500 crashes below the option's strike price, the investor exercises the put option to sell their shares at that guaranteed floor price, capping their downside. Conversely, if the equity value rises, the investor lets the option expire worthless, forfeiting only the premium paid, and captures the full upside of the rising stock market.

Study Resources:

ASX Options Education [↗\(https://www.asx.com.au/investors/investment-tools-and-resources/online-courses/options-course-quiz/introduction-to-options\)](https://www.asx.com.au/investors/investment-tools-and-resources/online-courses/options-course-quiz/introduction-to-options)

CBOE Learning Center – Options Basics [↗\(https://www.cboe.com/learncenter\)](https://www.cboe.com/learncenter)

CME Group Options Overview [↗\(https://www.cmegroup.com/education/courses/introduction-to-options/introduction-to-options\)](https://www.cmegroup.com/education/courses/introduction-to-options/introduction-to-options)